NYC Taxi Mobility: A Visual Story of Urban Movement

1. Introduction and Goal

Our project goal is to transform NYC taxi trip data (from 2015 to 2024) into an interactive, linked-view dashboard that answers the question: **How do taxi usage patterns vary across time, and what underlying factors drive these variations?** The dataset comprises Yellow Taxis, Green Taxis, and For-Hire Vehicles (FHVs), with temporal (e.g., weekly taxicab hires), spatial (e.g., start and end boroughs), and economic data (e.g., trip fares). Within our project, the goal is to build a **coordinated visualization system** where every interaction in one view propagates to all others, enabling users to explore the three axes of our problematic simultaneously:

- **Temporal dynamics:** How demand fluctuates across hours, days, weeks, months, and years
- **Trip characteristics:** How distance, fare, passenger count, and tipping behaviour interact
- **Behavioral patterns/anomalies:** What happens during disruptions like COVID-19, snowstorms in winter, and holidays

The site guides first-time users through a narrative that demonstrates its scale and recurring patterns, followed by the collapse and recovery resulting from specific disruptions, and finally a full, linked dashboard for free exploration. This two-mode design serves both casual (who follow the story) and expert users (who want to query the data directly) in the target audience.

2. Visualizations and Linked Interactions

The dashboard on our website will comprise five views that share a global filter state. When the user performs an action in any view (e.g., brushing, clicking, hovering, toggling), all other views update accordingly. Below, we detail each view, its interactions, and how it links to the others. The sketches and details about linked interactions are available in the appendix.

2.1 Visualization 1 (V1): Trip Volume Over Time (Stacked Area Chart)

This view, shown in Figure 1, represents monthly trip counts (2015 to 2024) as three stacked layers: Yellow, Green, and FHV, where months and trip counts are present in x and y-axes. One of the key observations is the COVID-19 collapse, which appears as a dramatic valley. **The interactions for this view would include:** Brushing-to-zoom on the x-axis to select a date range. This becomes the active temporal filter for the entire dashboard and converts from monthly to day-wise annotations; hovering shows a tooltip with exact counts for each taxi type, and scroll-triggered annotations appear as the user scrolls during the narrative introduction.

2.2 Visualization 2 (V2): Weekly Pulse Heatmap

This view, as heatmap, shows the hour of day for each day of the week, with cell colour intensity proportional to trip volume. It could reveal commute peaks (e.g., Monday to Friday at 8 AM and 6 PM), nightlife surges (Fridays and Saturdays after midnight), and relatively quiet Sunday mornings (as seen in the Milestone 1 EDA). **The interactions for this view would include:** Clicking a cell (e.g., “Tuesday 8 AM”) highlights all relevant data in V1 and filters V3 and V4 to trips from that time slot, hovering shows exact trip counts, and finally selecting pre-defined events from a dropdown for comparing rhythms across years.

2.3 Visualization 3 (V3): NYC Zone Choropleth Map

This will be a map of NYC’s taxi zone polygons coloured by pickup volume, as shown in Figure 1. We would use TopoJSON of NYC taxi zones from NYC Open Data, rendered with D3-geo. **The interactions for this view would include:** Clicking a zone (e.g., “JFK Airport”), which would filter all other views to that zone’s trips. A side panel would show some zone stats, such as average fare, average distance, peak hour, and top destinations. Hovering shows the zone name; toggling is possible between different years and between pre- and post-major events. We also plan to have a pan-and-zoom feature for spatial navigation.

2.5 Visualization 4 (V4): Trip Anatomy Explorer

Finally, as an exploratory tool, we want to create a scatter plot of fare vs. distance (from ~50K sampled trips), coloured by hour of day. This would reveal fare structures, tip outliers, and trip-type clusters (short Manhattan hops vs. long airport runs). Indeed, this is the view that directly addresses the "trip characteristics" axis of our problematic. **The interactions for this view would include:** brushing a region on the scatter (e.g., high-fare short-distance trips) converts V3 to highlight which zones from where those trips originate, and V2 shows when they happen, and axis selection that allows swapping axes to show fare vs. tip%, distance vs. duration, passenger count vs. fare, etc.

2.3 Visualization 5 (V5): Annotated Event Timeline (Scroll-Driven)

This view, shown in Figure 1, represents daily trip count as a line chart with vertical markers and callout boxes for major events (COVID lockdown Mar 2020, reopening phases, winter storms or blizzards, Thanksgiving and Christmas trends over the years, NYE dips, taxi fare hikes, etc.). Users will be able to select these trends, with details on how to filter and navigate to their desired time period. **The interactions for this view would include:** Click-triggered events auto-set the dashboard's time filter to the event's date range, and all views snap to that moment (shown in Figure 2); a readable step-wise description of the chosen filters to retrace the event steps.

Our core visualization of the entire project (MVP) comprises V1, V2, V3, V4, and V5 with at least one event story (e.g., COVID-19). We discuss the challenges and enhancements in Section 4.

3. Tools and Lectures

The project is implemented as a web-based interactive visualization using **D3.js** for rendering and interaction, combined with **Scrollama.js** for the narrative layer. It relies on standard web technologies (**HTML, CSS, JavaScript**) and DOM manipulation to dynamically update views.

Some basic tools introduced in the lectures are used directly to implement the system, including **DOM manipulation and event handling (Week 4)**, as well as **SVG-based rendering for visual elements (Week 4)**.

The design follows the lectures on **data types and visual encoding (Weeks 1–3)**, guiding the choice of charts, colour scales, and aggregation strategies. Interaction mechanisms such as brushing, filtering, and linked views are based on the **interaction lectures (Week 5)**. Finally, the scrollytelling structure is inspired by the **storytelling and design lectures (Weeks 7 and 12)**, enabling a transition from guided narrative to free exploration and browsing.

4. Enhancement ideas

Currently, the scatter plot (V4) introduces the **trip-characteristics axis**. V4 is more challenging than other views due to sampling constraints and cross-filtering with the map; therefore, we aim to first include a static version (pre-computed for certain time periods) and, as an enhancement, design the complete linked and dynamic version of V4. Other enhancements include smoother transitions between trigger events, particularly in V5. However, it might be possible since we could pre-compute the relevant transitions for the events.

Other advanced enhancements include alternative representations, such as parallel coordinates for multi-variable exploration (distance, fare, tip%, passenger count, duration), as well as focused sub-views, such as a tipping behaviour panel that highlights variations across zones, time, and fare ranges. These additions will enhance interpretability and analytical depth but entail higher implementation complexity. That said, we intend to implement some of these enhancements to improve the user experience and provide more insights into the dataset.

Appendix

A.1. Functional Prototype Status

Our current prototype includes:

- the website skeleton with the 4-panel dashboard layout and Scrollama scroll containers;
- V1 stacked area chart rendering with real aggregated data and working brush interaction;
- static panels for V2, V3, V4 with descriptive text;
- The global filter state bus is working so that brushing V1 already dispatches filter events;
- The complete data pipeline producing all required JSON aggregations from the NYC TLC Parquet files, with outputs hosted in the repository and on HuggingFace.

A.2 Global Controls

Besides the features and goals associated with per-view interactions (described in Section 2), the dashboard will have global controls that affect all views:

- **Taxi type toggle**: checkboxes for Yellow, Green, FHV.
- **Year range slider**: a slider for restricting the entire dashboard to a sub-period.
- **Reset button**: clears all filters and returns to the full-data overview.
- **Narrative mode toggle**: switches between scroll-driven story and free dashboard exploration.

A.3. Linked Interactions Between Views

The dashboard is designed as a coordinated system where all views share a common global state. Interactions in one view propagate to the others, enabling users to explore temporal, spatial, and behavioural dimensions simultaneously.

The main links between views are the following:

- **Temporal linking across visualizations:** brushing a time range in V1 (Trip Volume Over Time) updates V2 (Weekly Heatmap), V3 (Map), and V4 (Scatter) to reflect only that period. Similarly, scrolling or selecting an event in V3 sets the global time filter, updating all views accordingly. V2 also interacts with V1: selecting a specific time slot (e.g., Tuesday 8 AM) highlights the corresponding data in the timeline.
- **Spatial linking (V3 with V1, V2, V4):** clicking a zone in V4 (Map) filters all other views (V1, V2, V4) to trips originating from that zone. Conversely, selections in V4 (scatter brushing) highlight the zones in V3 corresponding to those trips.
- **Trip characteristic linking (V4 with V3, V2, V1):** Selecting a region in V4 (e.g., high-fare short-distance trips) highlights the corresponding zones in V3 and reveals when these trips occur in V2 and the trends in vehicle counts in V1.

A.4. Sketches for Visualizations

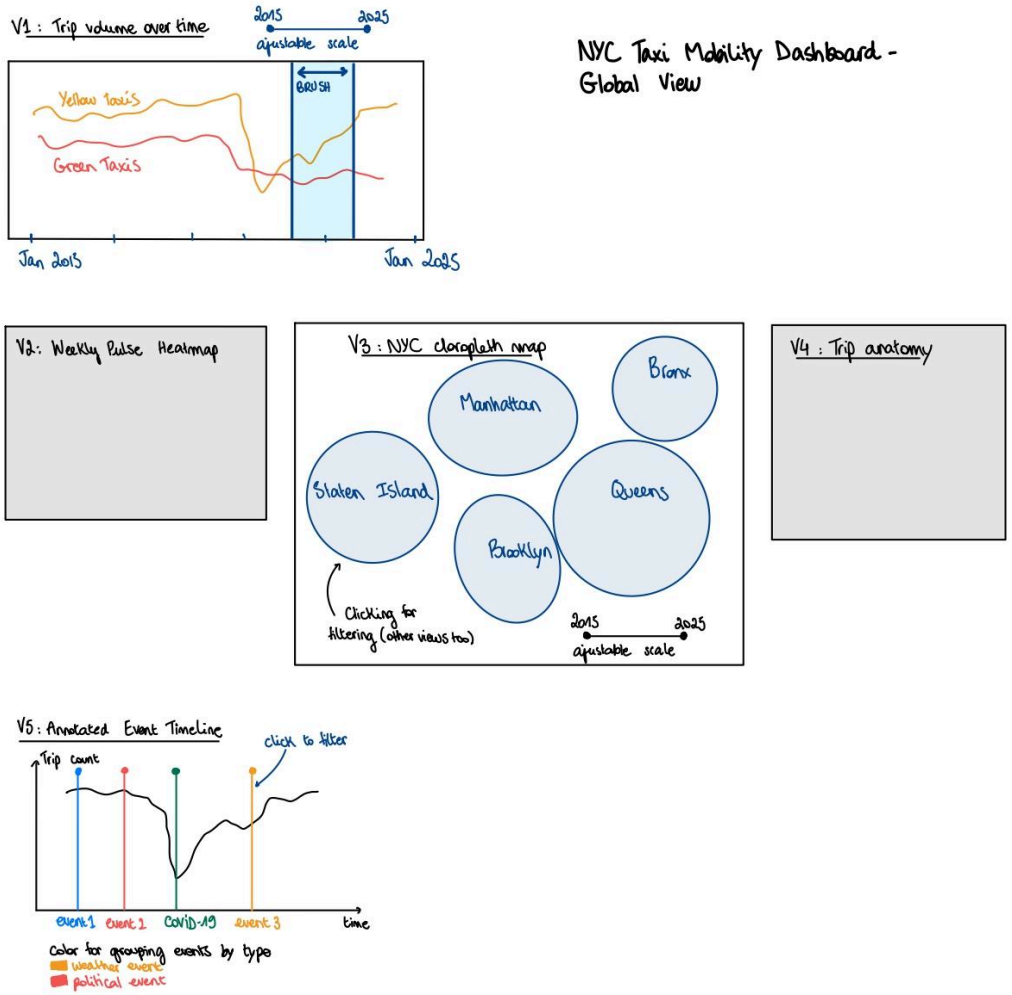

Figure 1 : V1-5 individual dashboard sketch

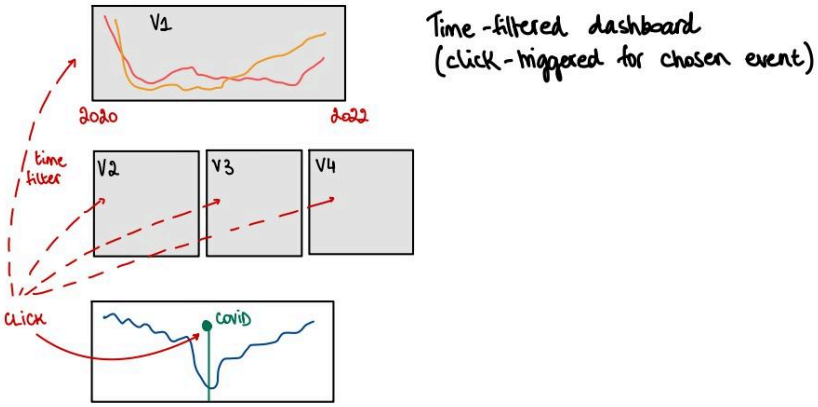

Figure 2 : Interactive linking dashboard Sketch